



Lesson 3.5 — Polynomial Arithmetic

Big idea: Adding, subtracting, and multiplying polynomials is just collecting like terms — it's arithmetic on the coefficients.

Quick review

Add / subtract: combine like terms (same variable AND same exponent).

Multiply: distribute every term in one polynomial across every term in the other. Multiply coefficients, add exponents on x .

Special products: $(a + b)^2 = a^2 + 2ab + b^2$. $(a - b)^2 = a^2 - 2ab + b^2$. $(a + b)(a - b) = a^2 - b^2$.

Closure: the sum, difference, or product of two polynomials is always a polynomial.

Worked example

Worked example. Multiply $(2x - 3)(x^2 + 4x - 5)$.

$$= 2x(x^2) + 2x(4x) + 2x(-5) - 3(x^2) - 3(4x) - 3(-5) = 2x^3 + 8x^2 - 10x - 3x^2 - 12x + 15 = 2x^3 + 5x^2 - 22x + 15.$$

Warm-ups

1. Add $(3x^2 - x + 2) + (x^2 + 4x - 5)$.
2. Subtract $(5x^3 - 2x + 1) - (x^3 + 3x - 4)$.
3. Multiply $(x + 5)(x - 5)$.

Practice

1. Multiply $(x + 3)(x^2 - 2x + 4)$.
2. Expand $(2x - 1)^3$.
3. Find a polynomial $P(x)$ such that $P(x) + (x^2 - 4x) = 3x^2 + x - 5$.

Challenge

1. Expand $(x - 2)^4$. (Use the binomial theorem or repeated multiplication.)
2. Show that $(x + a)(x + b)(x + c) = x^3 + (a + b + c)x^2 + (ab + bc + ca)x + abc$. Then use it to expand $(x + 1)(x + 2)(x + 3)$.



Answer key — Lesson 3.5

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $4x^2 + 3x - 3$.
2. $4x^3 - 5x + 5$.
3. $x^2 - 25$.

Practice

1. $x(x^2 - 2x + 4) + 3(x^2 - 2x + 4) = x^3 - 2x^2 + 4x + 3x^2 - 6x + 12 = x^3 + x^2 - 2x + 12$.
2. $(2x - 1)^2(2x - 1) = (4x^2 - 4x + 1)(2x - 1) = 8x^3 - 4x^2 - 8x^2 + 4x + 2x - 1 = 8x^3 - 12x^2 + 6x - 1$.
3. $P(x) = 2x^2 + 5x - 5$.

Challenge

1. $(x - 2)^2 = x^2 - 4x + 4$. Square again: $(x^2 - 4x + 4)^2$. Multiply: $x^4 - 8x^3 + 24x^2 - 32x + 16$.
2. Expand step-by-step: $(x+a)(x+b) = x^2 + (a+b)x + ab$; multiply by $(x + c)$ and group. Apply with $a=1$, $b=2$, $c=3$: $x^3 + 6x^2 + 11x + 6$.